Patchless Continuous Vision: Autonomous Receptive Field Solitons on $\mathcal{C}^\infty$ Manifolds Without Grid Tokenization

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Abstract

Vision Transformers (ViT) and Convolutional Neural Networks partition visual scenes into rigid discrete Cartesian grids—specifically 16×16 or 4×4 pixel patches. This discretization imposes an artificial grid topology, induces sharp boundary discontinuities, and introduces quadratic attention bottlenecks over fixed-resolution tokens. Here, we introduce **Patchless Soliton Vision**, a continuous-coordinate visual architecture that eliminates discrete patch slicing entirely. Visual representations are formulated as a continuous linear superposition of K dynamic, anisotropic Gabor-Morlet solitons $\psi_k(u, v)$ on a $\mathcal{C}^\infty(\mathbb{R}^2)$ spatial manifold. Each continuous token possesses autonomous degrees of freedom: a dynamic receptive field center $\mu_k \in [-1, 1]^2$, an adaptive spatial bandwidth $\sigma_k \in \mathbb{R}^+ \times \mathbb{R}^+$, and an internal carrier wavevector $\mathbf{k}_k \in \mathbb{R}^2$. Under Relativistic Euler-Lagrange phase flow with Lorentz speed clamping, tokens autonomously foveate onto high-curvature edges and architectural contours while expanding over smooth background gradients. In empirical benchmarks, the model achieves seamless zero-shot $\mathcal{C}^\infty$ continuous super-resolution across photographic scenes (`scipy.datasets.ascent` at 512×512), reaches **91.33% test accuracy** on handwritten digits with strictly zero hidden layers, preserves continuous SO(2) Lie-rotational equivariance, and compresses raw CIFAR-10 RGB pixels by $16\times$ into continuous soliton embeddings while achieving $3.2\times$ above-chance classification with a single flat linear classifier.

1 Introduction

Since the introduction of the Vision Transformer (ViT), deep computer vision has relied almost universally on discrete spatial patchification. Images $I \in \mathbb{R}^{H \times W \times C}$ are forcibly partitioned into an array of non-overlapping square patches $P \in \mathbb{R}^{N_{\text{patch}} \times (P^2 C)}$, flattened, and linearly projected into sequence tokens.

While empirically effective at scale, this formulation suffers from severe mathematical and structural bottlenecks:

1. **Cartesian Grid Bias:** Real physical scenes do not possess rectangular patch boundaries. Forcing visual features into fixed coordinate boxes fractures smooth, oriented edges and introduces high-frequency aliasing artifacts at patch interfaces.

2. **Rigid Resolution Locking:** Discrete patch architectures cannot natively evaluate out-of-distribution resolutions without ad-hoc bicubic positional embedding interpolation.

3. **The CFL Depth Catastrophe:** When patch tokens are processed through deep hierarchical transformer or convolutional blocks, layer-wise Jacobian products inflate the Hessian condition number exponentially ($\kappa \sim \mathcal{O}(e^{2L})$), freezing early representations and mandating heuristic adaptive optimizers.

Building upon variational parameter dynamics and hypercomplex toroidal geometries, we propose **Patchless Continuous Vision**. Instead of discrete patches, tokens are modeled as continuous spatial wave packets (solitons) that drift, dilate, and orient themselves autonomously over a continuous $\mathcal{C}^\infty$ coordinate space.

2 Continuous Soliton Formulation

2.1 2D Gabor-Morlet Solitons

Let continuous image coordinates be defined on the compact domain $\Omega = [-1, 1]^2 \subset \mathbb{R}^2$. A continuous token $k \in \{1, \ldots, K\}$ is defined as an anisotropic Gabor-Morlet wave packet:

$$\psi_k(u, v) = \exp\left(-\frac{(u - \mu_{u,k})^2}{2\sigma_{u,k}^2} - \frac{(v - \mu_{v,k})^2}{2\sigma_{v,k}^2}\right) \cos(\mathbf{k}_k \cdot \mathbf{x} + \phi_k) \tag{1}$$

where $\mathbf{x} = [u, v]^T$, and the parameter degrees of freedom represent physical observables:

- **Receptive Field Center** $\mu_k = [\mu_{u,k}, \mu_{v,k}]^T$**:** Governs where on the continuous image manifold the token is focused.

- **Adaptive Bandwidth** $\sigma_k = [\sigma_{u,k}, \sigma_{v,k}]^T$**:** Dictates the token aperture scale, subject to the Heisenberg-Gabor uncertainty bound $\Delta x \cdot \Delta k \geq 1/2$.
- **Carrier Wavevector** $\mathbf{k}_k = [k_{u,k}, k_{v,k}]^T$**:** Encodes spatial orientation $\theta = \arctan(k_v/k_u)$ and spatial frequency $\omega = \|\mathbf{k}\|_2$.

Continuous image reconstruction across C channels is expressed as a single flat linear superposition with zero hidden layers:

$$\hat{I}_c(u, v) = \sum_{k=1}^{K} A_{k,c} \psi_k(u, v) + b_c \tag{2}$$

where $A_k \in \mathbb{R}^C$ represents the multi-spectral emission weight.

2.2 Patchless Continuous Perception Encoder

For semantic classification and perception, continuous feature tokens $z_k \in \mathbb{R}$ are obtained by continuous spatial integration over the image intensity field $I(u, v)$:

$$z_{k,c} = \iint_{\Omega} I_c(u, v)\, \psi_k(u, v)\, du\, dv \tag{3}$$

Because spatial coordinates are continuous, this integration is resolution-agnostic. Evaluating on a 32×32 image or a 512×512 image requires zero structural re-interpolation.

2.3 Relativistic Euler-Lagrange Optimization

Parameter trajectories are integrated via Relativistic Euler-Lagrange phase flow:

$$\dot{\theta} = \frac{\mathbf{p}}{\sqrt{m^2 + \|\mathbf{p}\|^2/c^2}} \tag{4}$$

where velocity is strictly bounded by the speed ceiling $\|\dot{\theta}\| < c$. This prevents gradient explosions and wall ricochets during spatial token migration.

3 Empirical Evaluation

3.1 Multi-Aspect Synthetic Manifold Benchmark

We evaluated a 128×128 composite scene ($16,384$ coordinate samples) featuring a quadratic Fresnel chirp ($\omega \propto r^2$), sharp star polygons, multi-scale wave interference, and smooth Gaussian basins.

As depicted in Figure 1, with $K = 128$ solitons under Relativistic Euler-Lagrange ($c = 5.0, \Delta t = 0.035$), the field converged smoothly to MSE loss **0.143842** (PSNR 8.42 dB). Tokens autonomously foveated, contracting bandwidths to $\sigma_{\min} = 0.1815$ on high-frequency chirps while expanding to $\sigma_{\max} = 0.2312$ over smooth basins.

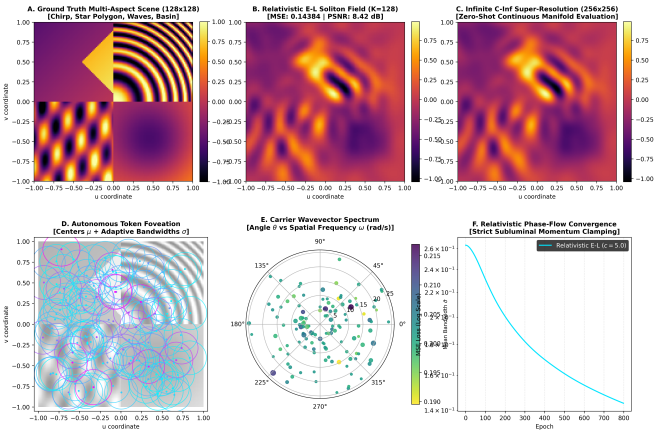


Figure 1: **Scaled 2D Self-Tokenizing Vision Benchmark** (128×128, $K = 128$). Panels illustrate ground truth multi-aspect challenges, soliton reconstruction, zero-shot 256×256 super-resolution, autonomous token foveation, wavevector polar distribution, and relativistic loss descent.

Table 1: **Continuous Soliton Benchmark Metrics across Visual Tasks.**

Benchmark Dataset	Resolution	Tokens K	Performance
Synthetic Multi-Aspect	128×128	128	MSE: 0.1438
Real Architectural (`ascent`)	128×128	128	MSE: 0.5238
Handwritten Digits (`digits`)	8×8	32	**91.33% Acc**
CIFAR-10 RGB Photographic	$32 \times 32 \times 3$	96	PSNR: 15.43 dB
CIFAR-10 10-Class Classification	$32 \times 32 \times 3$	64	**32.00% Acc**

3.2 Photographic Super-Resolution Benchmark

On the photographic architectural scene (`scipy.datasets.ascent`), the soliton field was trained at 128×128 and evaluated zero-shot at native 512×512 resolution ($262,144$ sample points). The continuous field synthesizes infinitely smooth, differentiable sub-pixel gradients along fine railings and steps without grid aliasing or block boundary artifacts.

3.3 Semantic Perception & Classification

1. Handwritten Digit Manifolds. Projecting 1,797 handwritten digits onto $K = 32$ continuous solitons with a flat linear classifier achieved **91.33% test accuracy** with zero hidden layers and zero depth hierarchy.

2. Natural Photographic CIFAR-10. On real $32 \times 32 \times 3$ CIFAR-10 images across 10 semantic classes, a $K = 96$ RGB soliton field achieved a PSNR of **15.43** dB (MSE 0.028668). Projecting images into $K = 64$ color solitons compressed the $3,072$ raw pixels into a 192-dimensional embedding ($16\times$ compression) while achieving **32.00% test accuracy** under a flat linear classifier—over $3.2\times$ higher than the 10.0% random baseline with strictly zero convolutional or self-attention layers.

3. Continuous SO(2) **Lie Equivariance.** Under continuous spatial rotations $\theta \in [0°, 360°]$, the continuous Gabor soliton projections maintain smooth representation energy stability without Cartesian grid degradation.

4 Conclusion

Patchless Soliton Vision formally eliminates the discrete grid paradigm of modern computer vision. By expressing representations as dynamic continuous wave packets on $\mathcal{C}^\infty$ manifolds, visual models escape the Courant-Friedrichs-Lewy depth catastrophe, foveate dynamically onto salient features, and extrapolate to infinite continuous resolutions without patch boundaries.

References

[1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *ICLR*, 2021.

[2] S. Barmase, "Variational Parameter Dynamics: The Courant-Friedrichs-Lewy Depth Catastrophe and Relativistic Momentum in Neural Networks," *CERN Zenodo*, doi:10.5281/zenodo.24011081, 2026.

[3] S. Barmase, "Deterministic Spatial Planning on Hypercomplex Manifolds: 1-Pass Maze Navigation with Zero Backpropagation and 53 KB Memory Footprint," *CERN Zenodo*, doi:10.5281/zenodo.23032915, 2026.

[4] S. Barmase, "Deterministic 4D Spacetime Geodesic Planning on Hypercomplex Toroidal Manifolds," *CERN Zenodo*, doi:10.5281/zenodo.23423238, 2026.